وزارة الاتصالات
وتكنولوجيا المعلومات

رواد مصر الرقمية

Customer Churn Prediction & Analysis Final Report

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Project Summary

Problem Definition

Customer churn -when a customer discontinues their relationship with a business is a significant issue in customer lifecycle management. The goal of this project is to build a machine learning model that predicts customer churn risk using behavioral and demographic features. Early prediction enables businesses to retain valuable customers and reduce revenue loss.

Data Exploration

The dataset consists of 23 columns capturing a wide variety of customer information:

- **Demographics:** age, gender, region_category

- **Account & Behavior:** membership_category, preferred_offer_types, medium_of_operation, internet_option , avg_transaction_value, avg_time_spent, avg_frequency_login_days, points_in_wallet

- **Engagement & Feedback**: used_special_discount, past_complaint, feedback, complaint_status

- **Target Variable:** churn_risk_score (binary: 0 = low churn risk, 1 = high churn risk)

Key Insights from EDA

1. Wallet Balance Matters Most

The feature **(points_in_wallet)** is the most influential predictor of churn, suggesting that customers with higher wallet balances are significantly less likely to churn.

2. Membership Category Strongly Predicts Churn

- Customers with **No Membership** have the highest churn risk (very high positive T-statistic).

- Premium, Platinum, Gold, and Silver memberships are associated with lower churn rates.

3. Customer Feedback is Highly Predictive

Several feedback types are statistically significant in explaining churn behavior:

- Negative feedback such as **Poor Website, Poor Product Quality, and Poor Customer Service** are strongly correlated with high churn.

- **Positive feedback** like User Friendly Website, Reasonable Price, and Quality Customer Care are linked with customer retention.

4. Login Behavior and Engagement

- **avg_frequency_login_days** and **avg_time_spent** are significant indicators. Customers who log in more frequently and spend more time tend to stay.

- **engagment_score** is also significant — more engaged customers are less likely to churn.

5. Transaction Behavior

- **avg_transaction_value** is significant: customers with higher spending are less likely to churn.

6. Referral & Offers Influence Churn

- Customers not referred by others are more likely to churn.

- Preferred offers such as Gift Vouchers/Coupons reduce churn risk, while customers preferring No offers are more likely to leave.

7. Device and Connectivity Channel

- Users operating via Smartphones and those with Wi-Fi connections show slightly higher retention than others.

8. Recency Metrics

- “recency_of_last_activity” and “days_since_last_login” are statistically significant: long inactivity periods are linked with higher churn risk.

Preprocessing Steps

- Missing values imputed

- Categorical features encoded using LabelEncoder or One-Hot Encoding

- Numerical features scaled using a standard scaler

2. Model Development

Multiple models were evaluated:

- Logistic Regression

- Random Forest

- XGBoost (final choice)

Champion Model: XGBoost

This model delivered the best performance on the test data:

- **Accuracy:** 95.09%

- **Precision:** 95.19%

- **Recall:** 94.94%

- **F1 Score:**95.04%

- **AUC:** 98.50%

Model Artifacts:

- Serialized using pickle

- Files: “best_xgb_model.pkl”, “scaler.pkl”

Feature Importance:

Important features included:

- points_in_wallet

- membership_category

- feedback

- avg_transaction_value

- avg_frequency_login_days

- engagment_score

3. Deployment Strategy

The project adopts a dual-deployment architecture for accessibility and scalability:

A. Streamlit Application

- **Purpose:** Business-friendly UI for churn prediction

- Features:

  - Real-time prediction on input data

  - Simple interface for non-technical users

- **Log File:"**streamlit_app.log"

B. FastAPI Application

- Purpose: Production-grade REST API

- Features:

  - predict endpoint with input validation

  - Swagger documentation at "docs"

  - Web UI for JSON-based predictions

- **Log File:**"fastapi_app.log"

Containerization

- Dockerized FastAPI app using a “Dockerfile”

- Deployed on port **9000**

- Entry: “uvicorn fastapi_app:app --reload --port 9000”

4. Business Implications

The ability to predict churn offers tangible business value:

- **Customer Retention:** Enables proactive intervention strategies

- Resource Allocation: Focus retention efforts on high-risk segments

- Revenue Impact: Reduces cost of customer loss and boosts loyalty

- **Product Feedback Loop:** Uses churn patterns to improve services

5. Key Insights, Challenges, and Strategic Decisions

Insights

- Behavioral traits (wallet balance, feedback, login frequency) are more predictive than demographic factors.

- Membership tier and preferred offers are strong churn signals.

- Negative feedback categories are consistently tied to high churn risk.

Challenges

- Handling missing values in “points_in_wallet” and “referral_id”

- Dropping uninformative or high-missing-rate features (e.g., “referral_id”)

- Ensuring consistent encoding and preprocessing during deployment

Decisions

- Selected XGBoost for its high performance and interpretability

- Designed a dual deployment strategy for business and technical users

- Applied Docker to ensure consistency in deployment environments

6. Future Enhancements

- **Model Monitoring** : Add drift detection and performance tracking in production

- **Versioning:** Use MLflow or DVC for model tracking

- **Auto-Retraining Pipelines:** Incorporate CI/CD for retraining with fresh data

- Cloud Deployment: Scale via AWS/GCP with load balancing

- A/B Testing: Compare multiple model versions in production